AGRITECH BUSINESS MODELS

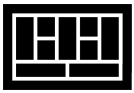


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SUPER GUIDE:
AGRITECH BUSINESS MODELS

BY DANIEL PEREIRA



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ISBN: 978-1-998892-21-1

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INTRODUCTION

Agriculture and agro-allied industries play an irreplaceable role in our lives and the development of our society as a whole. While many individuals only think about agricultural practices as a source of food, clothing, and other common items, they fail to understand that the agricultural revolution — which began about 12,000 years ago — is one of the most important discoveries of our species.

It catapulted us from hunter-gatherers scavenging off the land and left at the mercy of Mother Nature to shapers of the earth itself. It fueled the first major jump in the progress of our society, as more food meant more people. Also, as we got better at growing food we had more time to dabble in other disciplines such as science, the arts, and medicine, setting the stage for our future development.

In the present day, the impact of agriculture on our society has not diminished. In fact, with our growing population and ever-increasing demand for raw materials for various industries, agriculture seems to play a more pivotal role than ever before.

From an economic standpoint, the impact of agriculture on the economy of various nations and the world cannot be denied. However, this can vary widely between nations. Agriculture contributed to over 4% of the global gross domestic product (GDP) in 2018 according to a report from the World Bank, with this figure exceeding 25% of the GDP of certain countries with the category of the least developing nations.

To shed some further light on this disparity, let's take a look at

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the agricultural sector within different nations with vastly different backgrounds.

According to data from 2020, within the United States, 1.4% of U.S. employment comes from direct on-farm employment (farmers and ranchers), while this figure stands at 10.3% when the food industry is added. Within that same year, agriculture, food, and related industries contributed $1.055 trillion to the U.S. GDP, with direct on-farm activities contributing $134.7 billion.

Contrast this with a country such as the East African nation of Rwanda, where up to 72% of the working population are engaged in the agricultural sector, which constitutes about 33% of the nation's GDP — which stood at $10.33 billion in 2020.

This disparity also presents itself in an increasingly important part of the agricultural industry — technological innovation. Like most areas of human development, the agricultural sector has moved in leaps and bounds. From the simple farming practices of the first agricultural revolution to the mechanization of the industrial revolution to the mammoth industrial machines used in commercial farms today, innovation is a strong part of the agricultural industry. That’s why it comes as no surprise that it seems like the industry is making another slow transition to the next phase of its development: Agritech.

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WHAT IS AGRITECH?



An entire thesis could be written on the concept of Agritech — coined from the term agriculture technology —, but, in a bid to simplify the topic, it can be defined as

"THE APPLICATION OF MODERN TECHNOLOGY TO AGRICULTURAL PROCESSES IN A BID TO IMPROVE PRODUCTION, ENHANCE EFFICIENCY, AND STABILIZE THE AGRICULTURAL SUPPLY CHAIN".

From this deceptively simple definition, it can easily be seen that Agritech cuts across all areas within the agricultural industry and may represent the next major disruption within the sector itself. Even though the term technically includes more commonly known agricultural tools and machinery such as combine harvesters, produce processors, and even advanced irrigation systems, it is more commonly used to

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refer to what is referred to as the Fourth Agricultural Revolution.

WHAT IS THE FOURTH AGRICULTURAL REVOLUTION?

For a bit of context, it is arbitrarily assumed by most scholars and stakeholders that the agricultural industry has undergone three major revolutions so far.

The First Agricultural Revolution was the transition of our ancestors from a nomadic hunter-gatherer society to a more permanent farming society associated with planting and nurturing crops.

The Second Agricultural Revolution was the leap in the productivity and scale of farming through the mechanization of farm inputs. This not only allowed us to farm more land with fewer people, but it also opened up the industry to new markets with greater supply chains and transportation technology.

The Third Agricultural Revolution was the widespread adoption of species hybridization and genetic engineering of both farm animals and crops. It also involved the increased use of fertilizers and pesticides.

While all the previous transformations have involved technological advances, the Fourth Agricultural Revolution is seemingly centered around the mainstream adoption of Agritech into farming practices. This involves the use of technology such as drones, Internet of Things (IoT), robotics, data analysis, hydroponics, 3D printing, and much more.

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WHY DO WE NEED TO ADOPT AGRITECH?

The applications of Agritech are virtually limitless due to the constant increase in innovation and areas of possible application of this technology. However, it's also important to understand that the creation of Agritech stems directly from our limitations. Let's take a look at some of the key reasons why Agritech will represent the future of farming as we know it.

The Need for Improved Land Management Practices

Unsustainable land use practices have been a scourge that has plagued the agricultural industry since the very beginning. Poor land use practices such as monocropping, the overuse of synthetic fertilizers, pesticides, and herbicides, soil tillage, soil compaction, erosion, and soil contamination are just a few of the major land use issues associated with agriculture.

The adoption of more sustainable practices through the use of Agritech is a possible avenue for solving these challenges. Soil sensors, sustainable soil-improvement techniques, and the use of drone-aided soil analysis are just a few of the solutions the industry has presented already.

The Challenges of Climate Change

The key fact is the climate is changing. To what extent this change is due to human activities, how rapidly this transformation may occur, and what effects it may have on our environment in the future may be up to debate. Very few industries are as sensitive to the challenges posed by climate change as the agricultural industry.

It is also a well-known fact that agriculture contributes

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significantly to climate change through the release of greenhouse gases and the destruction of tree cover. Farmers all over the globe are coming face to face with the real-life effects of a changing climate in the form of weather disasters like droughts, floods, tornadoes, and organic disasters such as the increased spread of diseases among crops, animals, and even people.

Our Increasing Carbon Footprint

While every industry is guilty of increasing our carbon footprint on the planet, the agricultural industry plays an undeniably large role. Agricultural activities in the United States emitted an estimated 669.5 million metric tons of carbon dioxide equivalents in 2020.

This included the use of fossil-fuel-based synthetic fertilizers, the release of carbon sequestered in the soil, the burning of fossil fuels like coal, crude oil, and natural gas, as well as the release of methane, nitrogen, and CO2 through various processes.

Rapid Resource Depletion

Aside from being a mass producer of waste, agriculture is also a major consumer of resources like metals, petroleum, land, and other resources. This leads to the depletion of natural resources, adds to pollution, and worsens climate change.

Despite the vast resources available to us, without proper management and the push for renewable sources of energy and raw materials, we could quickly push ourselves towards the point of unsustainability. For example, studies have shown that over 100,000 hectares of land are lost each year due to unsustainable resource management, which is even more unsettling when you realize that one-third of the earth's land area is already unfit for crop growth.

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The Need for Greater Biodiversity

Of all the ways our activities affect natural biodiversity, the contribution of agriculture may be one of the most severe. Natural habitats such as forests, grasslands, and wetlands are frequently cleared to make way for new farmland and the population grows and demand surges.

This rapidly depletes the natural fauna and flora within the area, leading to consequences such as species extinction, global warming, ecosystem disruptions, and increased animal-transmitted diseases and attacks as we encroach ever closer to these natural habitats.

Water Management Hurdles

Despite the fact that over two-thirds of the planet's surface is covered in water, most of it is unfit for both human consumption and agricultural use. Only 3% of this water is freshwater, and only 1.2% of these freshwater resources are accessible to us.

Of this available 1.2%, about 70% is used for agricultural activities. This makes sustainable agricultural water resource management a top priority as many regions of the world battle with increasingly severe droughts due to climate change.

Our Population Is Growing Rapidly

The number of people on this planet is expected to hit 9.6 billion by the year 2050. With a study conducted by the United Nations stating that nearly one-in-three people did not have adequate access to food in 2020, it is important that we strive for innovative and sustainable solutions to the problem of rising food insecurity and a steadily growing population.

Sustainable practices enhanced genetic hybridization, and supply chain enhancements are just some of the answers to this challenge. We must look to Agritech as a source of

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further answers to this dire problem to avoid a possible humanitarian crisis.

USES OF AGRITECH

Despite the many challenges facing the agricultural industry, it's not all doom and gloom. Even as we speak, new innovative ideas are being tested out as possible solutions to the problems that we face, and sustainability practices are being more widely adopted all over the planet.

Agritech has contributed to this success as well in various ways, some of which have been highlighted below. Note that this is not an exhaustive list of the possible applications of Agritech, but just some interesting areas that we chose to highlight.

Smart Farming

Smart farming involves the use of proven scientific methods to increase farm productivity through greater farm produce yields, faster planting and harvesting, sustainable farming practices, and so on. In terms of Agritech, it usually involves the adoption of modern technology as well to facilitate these practices. Let's take a look at some well-known smart farming techniques involving Agritech.

- The use of smart irrigation systems which monitor the weather, soil conditions, plant water use, and evaporation to automatically optimize the watering schedule according to prevailing conditions. This offers better water resource management, energy savings, and environmental conservation than traditional irrigation systems which are programmed to simply turn on and off according to a preset schedule;

- The adoption of weather forecasting systems to optimize crop planting and harvesting is a great way to

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increase yield. This is due to the fact that many crops are very sensitive to prevailing weather conditions, and planting seedlings at the optimal time greatly increase their yield and conserves farm inputs as well;

- The use of sensors, drones, and satellites to facilitate precision farming systems allows the right crops and farmland to be accurately allocated to the right place and at the right time. It not only enhances yields and profits, but also supports sustainable farming practices such as mixed cropping;

- Hydroponic farming involves growing plants without the use of soil, which may be substituted with alternatives such as nutrient-rich water or various other suitable soil substitutes. Hydroponics has several advantages such as soil and water conservation, increased versatility in terms of location, the ability to be integrated into a vertical farming system, year-round production, and the fact that when done correctly it can significantly boost yields;

- Smart farming techniques have also led to the development of vertical farming, a system in which crops are grown in vertical layers through the use of innovative irrigation, lighting, and nutrient delivery systems. This is an effective way of growing crops in urban areas or regions which lack vast amounts of arable land.

Drones & Satellite – Scanning & Surveillance

Drone technology was one of the first Agritech innovations to see widespread usage. The use of drones has brought several key advantages to farmers, such as quicker and more efficient dispersal of farm inputs like seeds, pesticides, herbicides, and fertilizers, monitoring of crop and livestock health through visual and electromagnetic data such as thermal cameras and near-infrared (NIR) sensors, and even

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assist in plant pollination.

They are a vital part of precision farming and allow farmers to quickly map out large areas of land, estimate yield, detect early warning signs, and ultimately conserve energy through the use of more efficient, less fossil fuel-reliant survey methods.

IoT-based Sensors

Internet of Things (IoT) refers to the ability of various physical technologies to connect and communicate with each other through the use of sensors, software, processors, and other communication technology. While it is most commonly associated with concepts such as smart homes and smart cars, it is a very important part of Agritech as well.

IoT devices can be used in Agritech to detect, process, transmit, and monitor large amounts of data concerning crops, livestock, and prevailing weather conditions. This information can then be used to increase yield and save costs through the optimization of farm inputs. Some examples include IoT soil sensors, smart greenhouses, livestock monitoring technologies, smart delivery services, smart storage facilities, and so on.

Big-data

Despite its deceptively simple appearance, the average farm is an incredibly complex ecosystem, and minor changes in any of the numerous modifying variables can have a significant effect on yield. Due to this, even after the collection of this data from different sources, processing it into useful and actionable information had proven to be a challenge.

However, the rise of big-data analysis has allowed farmers and researchers to obtain a deeper and more holistic understanding of how a farm ecosystem works. This is done

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through the use of innovative technologies which allow this data to be effectively processed into simplified forms which aid in decision-making and allow a farmer to take simple steps to optimize yield. Big data has also been quite useful for crafting agricultural policies which affect food production and supply at both local and international levels.

Blockchain in Agricultural Logistics

Another component of the agricultural industry that has seen significant improvement due to the advent of Agritech is the area of supply chain logistics. Agricultural supply chains can be quite confusing to the average consumer and by the time the product makes it to the shelves or table, it is difficult to easily identify which exact farm it came from.

Certain key features which are native to blockchain technology have been leveraged to solve this issue by creating a system that allows both consumers and farmers to directly track the movement of farm products through the supply chain. This may also allow consumers to purchase products directly from particular farms, cutting out the middlemen and driving down costs, reducing their dependence on fiat currency, while at the same time giving them greater control of the items of the farm products that they consume.

Biotech

Biotechnology involves the utilization of cellular and biomolecular processes within organisms for the purpose of achieving various beneficial outcomes. In the context of Agritech, it involves the use of genetic modification and hybridization, vaccines, bio-pesticides and organic fertilizers, tissue culture, and so on. The aim is to improve both plant and animal yield, maximize profits, increase disease resistance, increase nutritional content, and reduce their dependence on fertilizers.

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Many agricultural startups within the biotechnology industry are mainly involved in the genetic modification of crops and livestock to improve certain qualities such as yield, improved soil fertility, crops that are resistant to adverse weather conditions, resistance against pests and disease, as well as shorter growing periods.

Farm Maintenance Technology

Increasing mechanization has long been the dominant green within the world of Agritech due to labor shortages as the number of people within the workforce involved in the agricultural sector shrinks. This has hit the industry hard, as agriculture is a particularly labor-intensive industry.

To combat this issue, significant investment has been made in the field of robotics and automation. The use of robots for certain tasks has helped ease the burden caused by a shrinking workforce and may even prove to be beneficial to farmers in the long run as robots have proven to be more efficient than human laborers in a number of tasks. However, the exorbitant costs associated with this technology still pose a major challenge to most farmers.

Production & Transportation Innovation

One of the major issues of Agriculture has always been transportation. A food mile is defined as the distance traveled by food as it moves from the producer to the final consumer. Innovators and entrepreneurs have been looking for creative solutions to cut down the number of food miles involved in a lot of agricultural supply chains.

Some of the solutions they have come up with include the adoption of hydroponic farms and vertical farms, which allow farms to be successfully supported in urban areas. Other solutions include digital marketplaces, and streamlined supply chains, which cut out the middleman and reduce the number of steps involved in the movement of a product from farm to plate.

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THE THREE TRENDS OF AGRITECH

One issue many stakeholders face when trying to categorize the different fields within Agritech is the versatile and usually overlapping nature of the technology within it. Therefore, in a bid to create a simplified paradigm of Agritech these technologies have been primarily divided into three:

- Technologies that enhance agricultural productivity through the use of innovation processes;
- Technologies that bring these products closer to the consumers through the use of more efficient supply chain technology and increased product availability;
- Technologies that are geared towards solving contemporary issues facing our society;
- Technologies that cut across industries and cannot be easily categorized.

TECHNOLOGIES THAT ENHANCE PRODUCTIVITY

Probably the early purpose for the integration of modern technologies into agriculture was for the purpose of enhancing yields and farm efficiency through the adoption of various innovative methods. Some promising examples of these include:

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- **The growing importance of desert agriculture**. As the population of the world rises and the availability of arable lands decreases due to increased urbanization, industrial consumption of raw materials, and poor land management practices, there is an increasing drive towards developing sustainable agricultural practices in regions that are traditionally difficult to cultivate such as arid areas or even deserts.

  This of course involves the use of innovative technologies such as efficient water reuse, saltwater desalination technology, drip irrigation, and genetic modification to create crops and livestock better suited to such environments. Already, many regions in the Middle East, North Africa, Australia, and even the United States have invested heavily in such technology;

- **The adoption of seawater farming**. As we mentioned earlier, only 3% of the world's water resources are fresh water, and only about 1.2% of this is accessible to use. Seeing that agriculture is the single largest consumer of fresh water on the planet, there is significant interest in the discovery of alternative sources of water for the purpose of agricultural activities.

  As arid regions battle with the unavailability of water, coastal regions are faced with the challenges of converting salt water into fresh water, as well as the conversion of the soil to one that is suitable for agriculture. Some of the technologies driving this push include desalination plants, seawater greenhouses, integrated farming practices that involve multi-species cultivation, and so on.

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TECHNOLOGIES THAT ENHANCE THE AVAILABILITY OF AGRICULTURAL PRODUCTS TO CONSUMERS

As the world moves further towards urbanization and rural-to-urban migration expands further in developing nations, it is becoming increasingly difficult to transport farm produce directly to consumers. Therefore, innovative supply chain technologies must be put in place to facilitate this process, either by enhanced supply chain efficiency, more efficient transportation and storage technologies, or even by finding creative ways to integrate agricultural activities within an urban environment. We have already mentioned some possible solutions such as vertical farming and hydroponics, but let's take a look at a few more examples.

- Greater interest in the widespread consumption of cultured meats. Even as we strive to increase available yields of available protein sources, alternative sources of sustenance must also be sought due to increased demand. One area which shows promise is the adoption of cultured meat, which refers to meat grown in a lab using animal cell cultures.

  Cultured meat may solve multiple issues such as the increased global demand for protein, food security due to the unreliability of animal and crop sources of protein with increased natural disasters and pestilence, animal welfare issues, ameliorating climate change and greenhouse gas emissions, and even human health. Though the industry is faced with challenges that include financial, technical, and ethical constraints, it does offer a promising alternative to

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farm-grown protein sources.

TECHNOLOGIES GEARED TOWARDS SOLVING CONTEMPORARY ISSUES FACING OUR SOCIETY

The agricultural sector may be a major contributor to climate change and environmental degradation, but it also has a significant role to play in the development of innovative solutions geared toward solving these crises. Let's take a look at a few areas in the agricultural industry has shown promise as viable solutions to many contemporary issues we face:

- **The adoption of algae feedstock as a potential biofuel of the future**. A feedstock refers to a raw material used to supply an industrial process or fuel a machine. For years, the growing concern surrounding the increasing utilization of fossil fuels and their deleterious effects on our environment has driven the development of alternative fuel sources such as biofuels.

  The use of microorganisms such as algae represents the third generation of biofuels and a shift from traditional substrates such as corn, sugarcane, soybeans, and so on. These could form the basis for sustainable fuel sources to drive various agricultural processes and may represent the future of the agricultural industry in the energy sector;

- **The use of bioplastics**. Bioplastics refer to a class of plastic materials that are produced from renewable

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biomass sources such as cornstarch, wood chips, vegetable oils, straw, recycled food waste, and so on. Many of these raw materials are sourced directly from the agricultural industry.

Some of these bioplastics are biodegradable, while others are not, and their ability to fit into existing recycling streams still remains in doubt. There is also some debate on whether bioplastics are actually more environmentally-friendly than traditional fossil-fuel-based plastics, they do offer a nearly infinite source of raw materials and allow us to drastically cut our dependence on fossil fuels.

TECHNOLOGIES THAT CUT ACROSS INDUSTRIES

Many of the newer technologies within Agritech can be applied to multiple industries and help in increasing yield, providing contemporary solutions, and increasing the availability of agricultural products. This makes them hard to strictly classify. They include concepts such as drone technology, IoT, precision agriculture, nanotechnology, data analytics, AI, food sharing and crowd-farming, and blockchain technology. Many of these concepts have already been discussed previously.

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FACTORS AFFECTING THE ADOPTION OF AGRICULTURAL TECHNOLOGY

Despite the obvious advantages of Agritech, there are numerous factors that may potentially stand in the way of its widespread adoption and further development. These include:

Institutional Challenges
While there have been significant leaps forward in the growth of agricultural technology in terms of policymaking and government support, more work needs to be done to drive the political will to support the further development of the industry. Policies that promote the adoption of these technologies by farmers and consumers through subsidization, grants, and concessions are several ways that policymakers can display that they fully support the implementation of Agritech solutions.

For example, political unrest in East Africa and a lack of political interest have long stifled previously promising research into desert farming practices. Also, the total or partial ban on genetically modified organisms (GMOs) by several countries all over the world and especially in Europe is a significant blow to the industry and one which may stifle further research.

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Economic Challenges

Even though the average cost of many of the technical inputs which make up the backbone of the Agritech industry has fallen in recent years, they remain exorbitantly high and out of the reach of most small- and medium-scale farms, especially in developing nations.

There are several ways this issue can be tackled. One is through both public and private sponsorship by relevant stockholders. Public and private grants, as well as subsidization, is also a viable channel. The adoption of economically feasible and sustainable Agritech practices is also very important, as farmers are business owners and will only adopt profitable practices.

Social Challenges

The increasing awareness of the dangers posed by climate change and our over-reliance on fossil fuels has been a major contributing factor in the push for the development of eco-friendly and environmentally sustainable agricultural practices such as hydroponics, vertical farming, and the use of renewable sources of energy.

However, certain Agritech practices have found it difficult to break into mainstream acceptance, a factor that has significantly hampered their development. Some prominent examples include the use of genetically modified organisms and cultured meat products. Also, certain social factors among the farmers themselves may affect their openness to adopting these novel solutions. This may include factors like age, gender, education level, social class, and so on.

Benefits Vs Cost

The feasibility of this technology must be carefully considered against its potential benefits. Running a farm is very similar to running any profit-oriented enterprise, and no sensible farmer interested in long-term sustainability will adopt any

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technology that pushes their business into unprofitability.

Therefore, research into innovative farming technologies should be focused on economically feasible options which can be adopted with minimal disruption. Innovation is also central in the process of driving down the cost of already existing technologies which may be currently out of the reach of the ordinary farmer.

Land Size

The effect of land size on the adoption of new agricultural technologies has shown mixed results. Smaller farms have a greater incentive to adopt newer, land-saving techniques which could potentially boost yield and profits, while the adoption of such methods on a larger scale may be out of the reach of many farmers.

However, the availability of significant areas of land is a prerequisite for the adoption of certain Agritech solutions. This makes such large-scale solutions not feasible for small landowners. This issue may be solved through shared farming practices, and there is still more research and innovation currently being undergone concerning this challenge.

Technical Issues

Despite farmer interest in the push for newer, more effective agricultural techniques and technologies, a significant number of the newer innovations associated with Agritech are highly technical and may be well beyond the ability of the average farm owner. This is a serious issue that may stand in the way of the widespread adoption of these technologies.

One way of overcoming this would be to hire the necessary human resources necessary to manage the technology. However, this will only further raise production costs and may end up transmitting these costs to the consumer, therefore

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defeating one of the primary purposes of adopting these newer technologies in the first place.

Information Dissemination

The ability to gain access to knowledge of these new technologies is also very important. This has to do with more than just the acquisition of knowledge of the existence of the technology, but also of its benefits, technical details, and how it can be adopted in their own particular situation.

Another important thing is how this information is perceived. If this knowledge is gained in certain situations it may create negative connotations concerning the technology due to various factors such as social issues, misinformation, unwillingness to change, and a situation where gaining knowledge about a particular topic only informs the farmer more on their ignorance of the topic leading to further negative associations with the technology. Therefore, it is important that this information is not only disseminated among relevant channels, but is disseminated in a manner that will encourage widespread adoption as well.

Local Factors

Crafting local solutions is also quite important due to the diverse and peculiar nature of the challenges facing farmers all over the globe. These difficulties may require drastically different solutions, even when they appear seemingly similar.

For example, desert agriculture technology designed for the arid regions of California in the Western United States may prove to be unfeasible in similar arid regions all around the world due to reasons ranging from cost to local climate conditions and even government policies.

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TECHNOLOGY INNOVATIONS SERVING AS THE FOUNDATION FOR NEW BUSINESS MODELS IN AGRICULTURE

The technologies which act as the driving force of the Agritech industry come from a wide range of fields such as biotechnology, mobile phones, AI, robotics and automation, blockchain technology, and so on. Let's take a deeper look at some of these technologies and the impacts they have had on Agritech business models, as well as take a look at a few agriculture businesses which adopt these technologies.

BIOTECHNOLOGY

Biotechnology involves the utilization of cellular and biomolecular processes within organisms for the purpose of achieving various beneficial outcomes. In the context of agriculture, it mostly involves the use of genetic modification, vaccines, bio-pesticides, organic fertilizers, tissue culture, molecular markers, and so on. The aim is to improve both plant and animal yield, increase disease resistance, increase nutritional content, and reduce their dependence on fertilizers.

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Agricultural Biotechnology Startups

Many agricultural startups within the biotechnology industry are mainly involved in the genetic modification of crops and livestock for the purpose of improving certain qualities such as yield, resistance to pests and disease, as well as shorter growing periods.

Genetic modification and hybridization are by no means new innovations within the world of agriculture, but newer technologies have pushed these techniques to new frontiers. Let's take a look at some of the companies behind the push for biotechnology within agriculture.

Microbe Plus

Microbe Plus is a Polish-based biotech firm founded in 2020 with the goal of successfully integrating biotechnology into Agritech and promoting sustainable agriculture through plant protection technologies. They specialize in the production of bio-based farm inputs such as pesticides and fertilizers.

These items have several key advantages. They are completely biodegradable and do not leave residues on the leaves or fruits of the plants. They can also allow farmers to choose a personalized selection of agro-products such as pesticides through DNA analysis of the associated pathogens, leading to more effective protection. They also select the optimal amount of fertilizer your crops need to thrive.

Bioheuris

Bioheuris is an Argentine company that was founded in the year 2016 that focuses on the development of herbicide-resistant crops. They do this through the integration of synthetic biology and gene modification to create plant strains that have this herbicide resistance by identifying the genes in resistant plants and swapping them into the crop DNA.

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They also help farmers fight herbicide resistance, an increasingly serious issue within the world of agriculture. Bioheuris does this by creating herbicide mixtures to reduce the chances of a plant becoming resistant to one particular strain. This also helps them reduce the dose of required herbicide, making the process more environmentally friendly.

Microendo

Microendo is another biotech firm that specializes in synthesizing personalized fertilizers that are safe for plants, consumers, and the environment. The Mexican startup was formed in 2019 and its process involves identifying the unique microorganisms associated with the healthy growth of that particular crop and using these microbes to craft an organic fertilizer.

Aside from just helping the crops, this also replenishes the natural microbiota of the soil and leads to the long-term sustainability of the farmland. They have several products which are targeted toward specific crops such as AgaveProtect®, CornProtect®, BerriesProtect®, and many other new products which are to be released soon.

ART Lab Solutions

ART Lab Solutions is an Australian biotechnology company started in 2018 which is involved in the breeding of improved cattle species, primarily through the use of in-vitro fertilization (IVF) technology. This involves the use of genetic research to create cattle breeds that offer higher yields, are more resistant to pests and diseases, and even produce more offspring.

Some of the technologies involved in this process include Ovum Pick Up-IVF (OPU-IVF) technology, the use of a serum-free high efficacy growth media system that is specially designed to meet the metabolic needs of the oocyte and embryo, as well as various technologies involved in

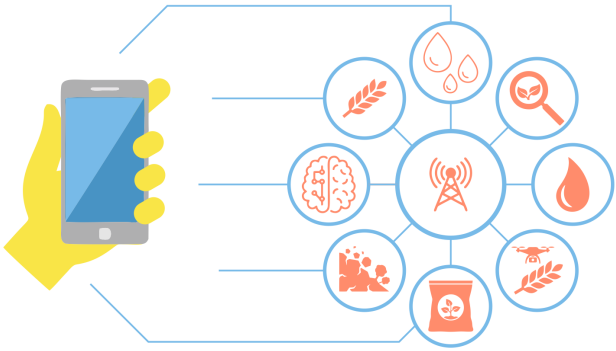
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mouse fertilization and embryo development for the purpose of genetic research.

BioPrime AgriSolutions
BioPrime AgriSolutions is an Indian-based biotechnology company involved in plant tissue culture and the use of agri-biologicals. The company makes use of various methods such as doubled haploidy, somatic embryogenesis, micropropagation, and in-vitro mutagenesis to support the breeding process.

To craft suitable agri-biological which are both safe for the plants and ensure maximum results, the company makes use of naturally-occurring plant compounds and plant-associated microbes. The basis of this is the symbiotic relationship formed in nature between many plants and the native microbial ecosystem. The first is also heavily invested in finding sustainable and long-term solutions to the issue of climate change, consultancy services, research and development, and more.

MOBILE PHONES



With the ubiquitous nature of mobile devices and their

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widespread integration into nearly every facet of our lives, it is of no surprise that they also form a key pillar of the Agritech business model. Mobile devices offer farmers all these services from a relatively cheap, portable, and widely accessible terminal.

Mobile phones have numerous applications yet some of the key uses associated with these devices within the context of Agritech include information dissemination and networking, remote monitoring through the use of IoT, and distribution channels for various financial services such as loans, insurance coverage, payment systems, and savings.

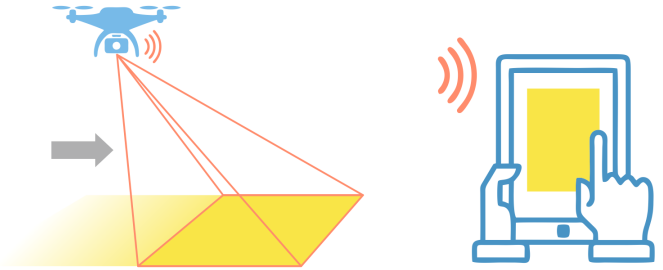
These devices can also be used for other purposes such as crop yield analysis through its integration with Big Data, disease diagnosis through the use of plant disease identification applications, access to information on the weather and climate, field and water resource monitoring, and so on.

INTERNET/CONNECTIVITY

As important as mobile devices are to various forms of Agritech technologies, internet connectivity is just as vital. Many of the technologies used within Agritech require some form of internet connectivity or local networks. This allows for remote activity, increased efficiency, global access to information, and so on.

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DATA ANALYTICS & AI/ML



Artificial Intelligence (AI) refers to the ability of machines to carry out activities or processes which would have normally required human intelligence, such as decision-making, visual perception, speech perception, and so on. Machine Learning (ML) is a type of AI that is focused primarily on building methods that allow machines to learn and improve performance through the analysis of available data.

Data analysis simply means the ability to detect, interpret, and communicate useful patterns found within available data. AI and ML have been playing an increasingly important role in this process and may be the answer to many of the problems facing agriculture.

Top Artificial Intelligence Startups Impacting Agriculture
Within the field of data analysis and AI, many startups are geared toward discovering feasible solutions to some of the issues facing agriculture such as resource management, how to increase productivity, and the safest ways to maximize profits. This can be done by helping the farmer make informed decisions or through full automation.

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AgEYE Technologies

Changing weather conditions, poor soil, diseases, weeds, and pests are just as few of the stresses that crops may encounter. A good farmer should be able to look out for these stressors and detect them early in order to affect changes that may save the crops or at the worst separate the unhealthy crops from the rest of the crops to prevent the spread of any potential stressors.

AgEYE Technologies is a US-backed startup that specializes in the production of AI technology that monitors plant stress through the use of sensors and alerts the farmer of changing conditions that may endanger their crops or how to optimize crop yield. This reduces the burden on the part of the farmer and helps them make informed decisions.

HelioPas AI

HelioPas AI is a German-based company that offers farmers advanced drought monitoring technology through the use of sensors that measure field data, weather conditions, and soil moisture. This allows them to accurately predict certain information such as periods of prolonged water shortage or less-than-average rainfall, the level, and quality of the available groundwater, as well as the soil moisture content.

Farmers can use all this information as an advanced drought warning system and a way to maximize their yield through well-informed irrigation practices and drought insurance planning. They also monitor situations of excess soil moisture, which are a significant risk factor for mildew.

Wolkus Technology Solutions

In many developing countries, there are two key problems facing the widespread use of AI technology as an integral part of the agricultural sector. One of them is the prohibitive cost of these technologies, another is the fact that many of

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the farmers in such regions farm on small-scale land holdings which is not ideally compatible with robotics automation.

Wolkus Technology Solutions is an Indian tech company that has offered several solutions to these challenges through its AI-powered platform Fasal. The platform offers affordable AI-powered sensors and monitoring technology which are designed for small-scale farms and can be monitored by the farmer through a variety of terminals.

OneSoil
OneSoil is a satellite data analytic company based in Belarus which offers farmers extensive satellite imaging data accessed directly from the European Union's (EU) Copernicus program. The Copernicus program is an initiative supported by the EU which offers Sentinel satellite data to the public.

By using this information, farmers can perform various activities such as allocate land boundaries, identify crops, monitor both cloud patterns and weather data, detect the ideal sowing date and phenophase of a crop, and so on. All this allows the farmer to make informed decisions while also saving time, cost, and labor.

Root AI
As farmers deal with increasingly unpredictable weather patterns, poor soil conditions, an increase in herbicide and pesticide resistance, as well as a higher crop disease burden, solutions such as indoor farming under a controlled ecosystem become more attractive. The issue is that most robotics and AI-monitoring technology is specifically designed for outdoor farming.

However, the US-based AI firm Root AI offers AI-based solutions specifically designed for indoor farming. These technologies do not only measure the soil and climate conditions within the indoor ecosystem, but also detect when

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crops ripen and help in the harvesting of these crops through the use of their patented robots which navigate the farm through the use of special indoor sensors.

IOT SOFTWARE AND HARDWARE

At this point, it is obvious that IoT has a central role to play in Agritech and its ability to solve many of the contemporary challenges facing modern agriculture. This is due to their versatility and the interoperability of the various sensors, robotics, and other forms of data-sharing hardware involved in IoT. All this information can be collated and accessed through common hardware like mobile devices, as long as the right software has been installed.

Top Internet of Things Startups Impacting Agriculture

The applications of IoT within the realm of agriculture are limitless. Let's take a look at some IoT forms that have made it their mission to discover solutions to agricultural challenges through the use of IoT and Agritech.

Xfarm

Xfarm is a Singapore-based IoT firm that deals with the use of agriculture-based IoT sensor technology to solve various agricultural issues. Their sensors receive information from the environment such as soil conditions, moisture content, humidity, sunshine, the presence of animal sounds, pests, and so on.

The system then processes this information into actionable data which allows the farmer to make informed decisions concerning his farm. This includes data visualization services and other analytic tools which include a historical logging feature as well. They help in the mapping and installation of the environmental sensors to ensure efficient placement.

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Mycroclime
Mycroclime is a Spanish company that offers a range of hardware and software Agritech IoT solutions aimed at environmental regulation, including cloud-computing services used to aid the integration of this technology. It was founded in 2016 and caters to greenhouses, hydroponic farms, and outdoor fields as well.

One of their most popular products is an array of temperature and humidity sensors that can be used to create a perfectly controlled greenhouse with a wide range of hardware designed for cooling, ventilation, and heating which the farmer can access from their mobile device.

Auroras
Auroras is an IoT-based Agritech firm that took IoT to a higher level of automation through their SAFE platform. SAFE is a special type of IoT array that does more than just detect environmental conditions and passes this information on to the farmer. It can act upon this information to make small changes to the environment through the use of an array of IoT hardware such as activated switches, motors, sirens, doors, curtains, fume extractors, and more to optimize these conditions.

This makes the SAFE system an almost fully automated wireless sensor and actuator network. It can be used in greenhouses, hydroponic farms, vineyards, and even outdoor fields as well.

Cultyvate
Cultyvate is a smart irrigation system based in India which uses a mix of IoT technology, AI, data analytics, and satellite imaging to create a fully automated irrigation system that helps farmers optimize water resource management on their farms. This is more efficient than traditional irrigation systems that simply work in a fixed schedule without taking into account the prevailing weather and soil conditions.

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The system operates through the use of satellite imaging, cameras, a system of sensors that measure both dynamic and static data, as well as an array of flow meters and valves. This ensures that crops receive only as much water as they need to optimize growth, especially when they need it the most.

Swinesmart

Animal husbandry is a more dynamic (and messy) form of agriculture than your typical crops-only farm. Animals wander, get lost or injured, fall sick, drop off productivity, and give birth to young. The process involved in getting animal farm products from the farm to the final consumer is certainly a long and complicated one.

That's why startups like Swinesmart make use of various IoT devices and sensors to help pig farmers closely monitor their animals and identify the optimal conditions which can be used to improve both animal safety and productivity. Their technology also allows animal products to be traced back to their farms of origin, and in certain cases back to the particular animal it was sourced from.

Lentera

Lentera is a Kenyan IoT firm that is dedicated to solving the problem of bringing Agritech technology to developing nations within Africa, where these technologies may be difficult to adopt due to their exorbitant prices or poor adaptability to native farming methods.

They offer a range of IoT sensors that monitor soil conditions, weather patterns, humidity, and the presence of pests in conjunction with satellite imaging technology. All this information can be made readily available to the farmer through the use of a mobile device.

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ROBOTICS AND AUTOMATION

The increasing integration of robotics and automation with AI technology and IoT may represent the future of Agritech and the agricultural industry as we know it. Now farmers are not only given the option of monitoring their farms but affecting the farm ecosystem remotely through the use of robotics. These systems have a wide range of applications within the sector such as transportation, seed, fertilizer, pesticide, herbicide dispersal, harvesting, monitoring, and so on.

Top Automation Startups Impacting The Agriculture Industry

Even though the agricultural sector makes up a significant portion of the GDP of most developed countries, both directly and indirectly, it is not a significant employer of labor. Chronic labor shortages within the industry may be one of the driving forces behind the strong push for greater automation within the agricultural sector.

Let's take a look at some robotics startups which have been making waves within the world of Agritech by creating sustainable agricultural solutions through the use of innovative automation technology.

Grofit

Grofit is an automation company that designs and markets a range of IoT devices custom-made for the agricultural sector. These sensors help farmers monitor various weather and soil conditions remotely through the use of specialized software. The sensors can detect conditions such as sunlight, humidity, temperature, evapotranspiration, soil moisture, and so on.

All this data is given to the farmer as actionable information, which allows them to make decisions concerning their farming practices based on the information available. Their

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data logging systems can store data for up to a month and all these services are available to farmers through mobile applications which can be downloaded on any internet-enabled device.

IgnisNova

IgnisNova Robotics is a robotics and automation firm based in India which offers IoT and robotics solutions to farmers through the use of AI-powered systems. These remote IoT sensors allow the farmer to easily access an entire world of information using on-site sensors and satellite imaging to obtain information on environmental and soil conditions, as well as crop health and weather patterns.

This information can then be integrated with various robotics and smart systems on the farm to optimize conditions in ways that improve crop growth and ultimately increase yield and efficiency, while at the same time driving down costs for both the farmers and consumers.

Deepgreen

Early crop disease detection systems are another area that has been receiving a significant amount of attention within the field of Agritech. They are built not only on IoT sensor technology but also advanced visual recognition systems which are powered by AIs and allow farmers to detect diseased crops early and separate them from healthy plants.

Deepgreen is an AI and robotics company that merges the fields of IoT, Ai, robotics, and automation to create a catalog of technological solutions which allow farmers to remotely monitor crop health and through the use of Application Programming Interfaces (APIs) that integrate efficiently with various agricultural hardware, sensors, tractors, and robots to carry out various labor-intensive farming operations.

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Sensonomic

Simulation technology is a well-researched field within AI, yet it has not seen much use within the world of agriculture. However, a Norwegian company known as Sensonomic has plans of changing this status quo by introducing a line of Agent-Based Modelling (ABM) simulations that can be used to predict anything from crop growth to weather models and even on-field farm operations.

The advantage of this is that it allows farmers within valuable information which can be used to plan farming activities and accurately predict the effects of various activities and weather patterns on crop growth. The company encourages users to be involved in the design process to create a uniquely adapted model which can be used for things like harvest planning, resource management, supply chain logistics, and risk mitigation.

Sagri

As the future of farming moves closer and closer toward full automation, it is important that such a system is supported by reliable software which can be integrated into existing systems and can be utilized effectively by the average farmer. The Japanese company Sagri is involved in the design of such software and specializes in the creation of smart farm management software.

This software is designed to integrate IoT sensors, satellite devices, robotic equipment, and crop data into a farmer-friendly AI interface that provides actionable data which can be used to optimize farm yield and conserve valuable resources.

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BLOCKCHAIN / DLT

The adoption of blockchain technology within the framework of Agritech has long been seen as one of the most revolutionary applications of the technology. Blockchain may be the key to revolutionizing the traceability industry through the use of unique identifiers to track agricultural products as they move from the farm to the consumers, ensuring safety and enhancing supply chain efficiency.

It may also offer a way for farmers to sell their goods at lower prices by eliminating middlemen and creating a decentralized marketplace where buyers and sellers can interact directly. Blockchain technology could also be used within the insurance industry to easily obtain information about the location, safety, and compliance information of any particular good.

Blockchain Startups Transforming The Agricultural Industry
Let's take a quick look at some startups which have been transforming the agricultural sector through the use of blockchain technology.

WorldCover
WorldCover is an insurance startup that was founded by various MIT computer science and finance professionals and is geared towards building an insurance platform to address the significant climate risks facing the world today. Every year, farmers lose billions of dollars worth of produce due to the direct and indirect effects of climate change on the environment. This is especially important for developing nations, as 70% of the 2.5 billion living at the base of the income ladder directly rely on agriculture for sustenance and income.

By working through a system of satellites, WorldCover

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monitors rainfall and automatically triggers payouts to farmers under their coverage when rainfall falls to levels that may endanger crop yield. This helps cushion the economic impact of natural disasters such as drought.

TE-FOOD
TE-FOOD is a traceability startup that makes use of blockchain technology and unique identification tools to track various forms of agricultural products throughout its supply chain. The company issues this information for various purposes such as improving operational efficiency by streamlining supply chain logistics, providing proofs of product safety and quality sustainability certifications, supporting product exports, and allowing for the quick withdrawal of products from the market.

Ripe.io
Ripe.io utilizes blockchain technology to build a transparent digital food supply chain in order to foster long-lasting consumer confidence. The startup creates the Blockchain of Food by mapping the journey of agricultural products along the supply chain through the use of high-quality food data.

Etherisc
Etherisc is a blockchain-based insurance firm that offers farmers crop insurance coverage through the use of decentralized insurance applications. This works through an extensive decentralized weather monitoring service that broadcasts weather information to both parties from different reliable external weather data sources and triggers insurance payouts immediately when an extreme weather event that may affect crop yield is detected. This reduces the risk of tampering on the insurer's end.

Demeter
Named after the Greek goddess of agriculture, Demeter is a blockchain startup that allows farmers to easily lease and

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farm micro fields anywhere in the world without the need for middlemen, unnecessary complexity, or the overhead of a big organization which would only drive up costs without necessarily adding value to the service.

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WHAT IS AN AGRITECH BUSINESS MODEL?

An Agritech business model is simply the operational plan of a business organization that is a part of the Agritech industry. A business model is expected to outline how a business enterprise operates, its products/services, its monetization model, expected costs, customer base, and other details.

Within the world of Agritech, there are dozens of business models that potential entrepreneurs can successfully adopt. These models can be successfully adapted to both large-scale and small-scale farm holdings and may be involved in the service industry, research and development, financial services, engineering, and so on.

THE RAPID EMERGENCE OF AGRICULTURE TECHNOLOGY (AGRITECH) BUSINESS MODELS

One of the key success stories of the Agritech industry has been its rapid rise in recent years and the steady adoption of Agritech technologies among even small-scale farmers. There are several reasons for the rapid emergence of Agritech business models in recent years.

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Increased Digitization

At the very backbone of the Agritech industry is the increased digitization of the modern world. Even within rural communities and developing nations, you would be hard-pressed to find regions without access to a mobile phone with internet connectivity. This simple fact had many beneficial effects on farmers and Agritech alike.

First of all, this increased digitization and interconnectivity made farmers much easier to reach by Agritech companies. It opened them up to a whole new world of information and opportunities by breaking down the barriers of data asymmetry. With this increased communication and transparency, this previously untouched market segment could become valuable members of the Agritech value chain.

Another advantage of this rapid digitization is that it opened up a feasible integration channel between farmers and Agritech companies. Many farmers do not have access to high-tech computers or hardware, but many of the programs used by Agritech companies can be accessed using simple applications of SaaS programs, which can be accessed through your mobile device. For example, information on weather forecasts can be shared easily among farmers or crop prices can be closely monitored using blockchain technology.

A Shift in Investor Interest

As Agritech sees increasing innovation and newer methods are discovered, more investors pick up interest in the industry as a potential investment opportunity. One reason for this may be due to the fact that purchasing farmland is becoming an increasingly low volatility investment as advances in technology allow us to better understand and adapt to our changing climate.

Another reason is the growing demand for farm products as

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the population grows ever larger and increasing urbanization sees a significant number of traditional agricultural economies leave the agricultural industry, leaving room for new investors.

The growing number of possible financial services associated with the agricultural and Agritech industry is also quite important. Investors with little or no background in agriculture can now successfully invest in farms through the use of different financial products such as farming-focused real estate investment trusts (REIT), agricultural stocks, Agricultural Exchange Traded Funds (Ag ETFs), Agricultural Mutual Funds, and so on.

Increased Accessibility To New Technologies

One of the major challenges the Agritech industry faced early on was the exorbitant prices of the technology associated with their use. However, rapid innovation and government subsidies have to a large extent driven down the prices of many of these technologies to levels that more farmers can afford. For example, the prices of solar panels have fallen by over 80% in the last decade, offering farmers a more affordable source of renewable energy and a chance to reduce their reliance on fossil fuels.

Also, the price of IoT sensors which are another vital component of many Agritech solutions has fallen by over 50% in the last decade as well. Combine this with increasing digitization and suddenly farmers have a whole new range of options open to them in terms of Agritech solutions. Further investment in Agritech drives profits within the industry, leading to more innovation and even cheaper technologies, creating a positive feedback cycle that benefits both parties.

Changing Political and Social Sentiments

Also of note is the changing political will towards Agritech and the push for newer innovative technologies within the

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agricultural industry, which has largely been driven by a shifting social sentiment concerning climate change and over-dependence on fossil fuels. As societies all around the globe realize the growing need for cleaner energy and more sustainable farming practices, the Agritech industry is pushed even harder to offer solutions to these challenges through the integration of modern technology and agriculture.

Some of the consequences of this change in sentiment are increased government subsidy programs for farmers and increased regulation concerning the use of fossil fuels, synthetic fertilizers, and pesticides, the use of renewable energy, sustainable farm practices, and eco-friendly waste management systems. It has also driven the private sector to invest in Agritech as consumer sentiment shifts towards greener farming practices and the demand for organic products rises.

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AGRITECH BUSINESS MODELS

As we mentioned earlier, there are numerous Agritech business models which exist within the many fields of Agritech. Let's take a look at some emerging models which have been found to be quite popular.

FARMER ADVISORY/CONSULTING

This is a digital advisory service whereby Agritech firms offer expert advice and consultation services to farmers, typically over digital platforms. The farmers gain important insights which can help them optimize their profits by adopting Agritech solutions that increase yield and efficiency while at the same time driving down costs in the medium- to long-term.

This can be especially helpful for smaller farms due to the fact that they typically operate on much thinner margins than larger commercial farms and therefore need all the support they can get. Some of the valuable information offered by these consultation services includes things like pest identification, seed selection, agronomic advice, climate prediction and information, planting timing, and so on.

There are several advantages to this type of business model. One is the ease at which this information can be disseminated through the use of mobile devices and the internet, even social media platforms as well. Another is the

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fact that after the initial cost of successfully setting up the platform, further operating costs are low since each additional farmer that is added to the system costs the business very little. Also, once trust is built between the advisory firm and the farmers there is a potential for rapid growth since they will likely suggest the service to other farmers, helping better position the company's brand image in the industry.

Despite this, there are also several challenges facing this type of system, especially when dealing with small-scale farms. One is that monetization of the service may be difficult, especially using a subscription-based monetization model. Another monetization option may be selling the farmer data to other businesses through a B2B mode. This has been found to be a more reliable option.

Thus, most advisory services offer a free package and then monetize their customer's data and sell it to other businesses. At the same time, they may offer a paid premium package that offers additional services.

PEER-TO-PEER LENDING

Sourcing financial support for their agricultural inputs and activities has always been a challenge for farmers, especially small-scale farmers. This is because the loan market among such farmers is typically informal and involves the use of local loaning agencies which usually charge exorbitant interest rates which may be as high as 10% per month in some cases. This makes repayment quite difficult for farmers, since most farms of such size operate on very thin margins of profit.

However, there are several compelling reasons for the type of loan system seen in this situation and the high-interest rate they charge. The first is that large-scale banking institutions find it hard to adequately penetrate the small-scale farming

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sector, especially in developing nations. This may be due to reasons like:

- The high level of inherent risk in small-scale agriculture discourages many banking institutions;
- The lack of knowledge that banks have or are able to accurately obtain on the individual credit worthiness of each farmer;
- Poor acceptance of the Know Your Customer (KYC) process among farmers, which usually involves a substantial amount of paperwork;
- The significant cost of customer acquisition due to the remote location and nature of such small-scale farms.

Due to these reasons and more, most farmers would rather choose local loan agents even if that means higher interest rates, and most banks do not consider the customer acquisition costs (CAC) to be worth their lifetime value (LTV).

However, digital lending platforms may offer the solution to some of these core issues. One popular form of digital lending which seems to be making waves is what is known as Peer-2-Peer (P2P) lending. A P2P lending platform is a digital financial service that allows farmers to obtain small-scale loans from interested investors, who typically charge a lower interest rate than the local loan agencies. P2P platforms function as intermediaries between these two groups and are typically paid a commission for this service.

These farmers typically obtain these loans at the start of the planting season and repay at the end, favoring short-cycle crops since they provide a quicker return on investment. These loans may be given as vouchers instead of cash to reduce the incidence of fraudulent activities and reduce the rate of default. However, even this form of digital payment is

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plagued with certain difficulties as well, such as the inherent risk of small-scale farming, the dangers of non-performing loans, the poorly regulated nature of the industry, pricing systems and rates, and so on.

TRACEABILITY

Changing consumer sentiment has driven more accountability within the agricultural and food industry, with a growing number of consumers taking an active interest in the source of the agricultural products they purchase. This may be for safety reasons such as fear of the potential contamination with dangerous pesticides, herbicides, and microorganisms.

It may also be due to social or moral issues such as the increased drive for cruelty-free animal products, organically grown crops, the push against GMOs, the drive for more sustainable farming practices, and the growing awareness of the presence of forced labor within the agricultural sector in many developing nations.

Due to this, many consumers and even farmers themselves are pushing for greater traceability, transparency, and easy access to information concerning the source of agricultural products and the farming practices involved in their production. This has led to the rise of the traceability industry.

The traceability business model is built upon the incentive of farmers, agro-allied industries, and primarily the food industry to ensure their products meet the standards set by various private and public or internal and external regulatory bodies due to growing consumer interest in these certifications. Therefore, a company that fails to source its agricultural raw materials from a farm that meets these criteria is bound to encounter significant backlash from its consumer base, which

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weakens its brand image and competitive advantage.

Many businesses have set themselves up as traceability firms that use various forms of inventory software, IoT sensors, databases, and blockchain technology to trace agricultural products or raw materials all the way back to the farm of origin. The industry includes the farmers, the firms which source agricultural raw materials from them, consumers, various certification bodies which set the standards, field mapping and survey teams, farmer training, and other relevant stakeholders.

There are different ways this business model may appear, but broadly there are two commoner forms. This may include simply creating a “white list” of certified farms that meet set standards and from which manufacturers are encouraged to source agricultural raw materials from. It may also involve a more complex process known as batch traceability, where all the agricultural products are given unique identifiers which are maintained throughout their existence in the supply chain and allow them to be accurately tracked as they go from “farm to plate”.

Traceability is primarily a B2B business model, as the incentive to maintain these standards rests not primarily on the consumers, but on the brands from which they source their goods. There is still significant development going on concerning the potential monetization method of such a model, but one important emerging trend is the commercialization of farmer data and the sale of this information to agro-allied firms and financial institutions. Also, farmers and businesses which source their raw materials may pay for a certification survey in order to boost their competitive advantage over other businesses.

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DIGITAL MARKETPLACES

To fully understand the appeal of this business model, it is important to understand the paradigm shift that occurred due to the widespread adoption of e-commerce and the digitization of the marketplace between buyers and sellers. The rise of e-commerce had several key advantages, such as bringing consumers and sellers much closer, thereby cutting costs by getting rid of the middlemen.

It also exposed both parties to a much wider market than they would previously have been able to access through the use of internet technology and online catalogs, vastly increasing sales volume. They save both merchants and buyers costs due to the removal of unnecessary physical warehouses and the virtually limitless shelf space offered by online e-commerce.

With respect to agriculture, small-scale farmers may be able to utilize this digital market for both the purpose of buying and selling. For example, purchasing farming inputs from an online marketplace would allow farmers to purchase these goods at a cheaper price due to a less fragmented supply chain. Also, they would be able to leverage the key advantages of e-commerce to sell their goods to a much wider range of customers.

Various e-commerce platforms could facilitate this process by acting as intermediaries between buyers and sellers for a relatively small commission. They will offer the farmers limitless shelf space, access to a wide range of buyers, and streamlined payment services.

However, there are several key challenges that may stand in the way of the widespread adoption of such a business model within the agricultural sector. One of the key

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challenges they face is farmer literacy concerning the use of digital marketplace platforms and technology. Significant efforts will have to be made to educate farmers on the advantages of such platforms and how to integrate these technologies into their current business model.

Another issue the industry may face is the intrinsic nature of the farming industry. Most farmers sell their produce at wholesale volumes, while most consumers are mostly retail purchasers. Also, these retailers may want to purchase a variety of items, while most farmers specialize in the sale of relatively few agricultural products.

The answer to this issue may be the emergence of a digitally enabled trader who maintains an inventory of goods which allows them to aggregate enough products from each farmer but still utilize the advantages of a digital marketplace to sell these products to individual consumers. These online traders would provide auctioning services, logistics and product tracking, streamlined payment services, and much more.

MECHANIZATION, SYSTEM LEASING, AND RENTALS

As we mentioned before, one of the major problems facing modern agriculture is chronic labor shortages the industry faces as the rate of rural to urban migration steadily rises. Since agriculture is a very labor-intensive occupation, this leaves many farmers with the challenge of finding ways to attract labor and maintain optimal productivity. This has led to an increase in labor costs as farmers have to offer higher day rates to attract willing workers, leading to a rise in the cost of farm products as well.

One of the solutions to this issue is the increased

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mechanization of farm labor through the use of robotics and automation. These devices can carry out farm operations with little or no human supervision, and in many cases are even more efficient than human labor. Though the initial investment in such machines may be hard, especially for small-scale farmers, these devices typically offer the farmer greater profits in the long run through greater efficiency and higher yields.

IoT sensors, robots and automation, digital connectivity, smart farming, and precision farming are just a few of the key areas involved in the development of agricultural mechanization, yet they form the backbone of the industry. Though, another key area has been emerging within the market which may be a solution to the challenge of employing mechanized agriculture on small-scale farms. The concept is referred to as a digital Mechanization Platform.

Even though the mechanization of farming may offer farmers greater returns, in the long run, the initial costs are much higher than most small-scale farmers can afford. A solution to this that many farmers employ is leasing these inputs for a specified period of time instead of purchasing them. However, the traditional system of leasing farm machines is often expensive and filled with inefficiencies.

By offering farmers a digital platform where they can apply for the right to temporarily lease these inputs from the owners, allowing them to make use of this equipment without having to worry about things like maintenance costs, storage, insurance, and so on depending on what was stipulated within the Service Level Agreement (SLA).

In certain situations, farmers may retain ownership of these inputs, depending on their agreement with the lender. This system also allows both the equipment owners and farmers to save costs through more efficient scheduling services by

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allowing the owners of these inputs to take advantage of clusters of demand.

For example, if several farmers in a particular region want to make use of equipment it is much cheaper for both parties for a particular individual to supply the equipment to those farmers as it cuts down transportation costs and ensures the equipment is being constantly utilized allowing them to recoup their initial investment which went into purchasing the machine.

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SOME OTHER AGRITECH BUSINESS MODELS TO LOOK AT

Here are a few more key Agritech business models that entrepreneurs can pursue.

ENGINEERING SERVICES

Within the context of Agritech, an engineering services model offers customers problem-oriented solutions to technical issues they may have concerning agricultural practices. This may include mechanical, electrical, software, and electronic engineering. Some examples include designing greenhouses, smart irrigation, and water recycling systems, using software to integrate various IoT devices on the farm, and so on.

Engineering businesses may offer a single technological solution — for example, the installation of solar systems — or may be multidisciplinary. A multidisciplinary model exposes you to a wider client base and may act as a hedge against fluctuations in demand.

A key part of this business model is your ability to adequately walk your client through the entire process, starting from what solutions you offer and how they stand to benefit from them, the survey and design phase, as well as construction, commission, implementation, and maintenance of the system. Your clientele may be large-scale or small-scale farms, so it's

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important that you have a catalog of different options which are suited for different situations.

There are several key resources that every good engineering business model must capitalize on. First is constant access to updated and relevant information concerning the industry and the ability to leverage this information to create uniquely tailored solutions for your clients. Other important resources include access to a range of effective tools and equipment, skilled labor, parts, and a network of engineers in allied fields who can be hired to consult or help on projects.

It's important to accurately plan out your budget beforehand and factor in some of the common costs which go into each project such as permits, labor costs, the costs of acquiring parts and tools, and so on. Scheduling is another big part of the business model, as project delays may significantly impact both your client and yourself in various ways.

MANUFACTURING

Manufacturing within the world of Agritech is quite important because they supply the tools, machines, and devices that are used by virtually every sector of the Agritech industry. There are numerous possible business models within the manufacturing industry, but, before venturing into the sector, it is important to carefully outline your value chain.

This includes which item (s) you intend on manufacturing and for which industry, are these items considered high-tech or low-tech, are you going to use raw materials or pre-fabricated items, are you producing the final end-products, or just another step along the production chain? Only after carefully answering these questions can you move forward.

Also, you must consider your customers as well. Most

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manufacturers within the Agritech industry usually deal with other businesses, so you must employ an effective B2B model, which involves understanding all the unique challenges and characteristics of the model as well. You should also be aware of whether your market is local or export, with local markets being cheaper, less complicated due to the fewer regulatory and bureaucratic steps, and better for new startups. However, export markets offer rapid scalability and may be an opportunity to expose your products to a much wider market.

Operations within a manufacturing company can vary widely depending on the unique product offered by the firm, its customers, and the market condition. Generally speaking, it involves logistics such as raw material and equipment procurement, supply chain management, production capacity, production type, environmental impact and sustainability practices, waste management, market analysis, energy efficiency, customer service, and so on.

PROCESSING AND TREATMENT

The agricultural industry is a primary industry and as such produces raw materials which must be further processed into secondary goods before they can be sold to the final consumer. Also, it is an industry that produces a significant amount of waste as well, which may require further treatment into forms that can be safely and easily disposed of without causing harm to the environment or people. Therefore, the processing and treatment industries form an integral part of Agritech and the agricultural sector as a whole.

These industries often process these raw inputs into products that have a higher resale value than the original items depending on market conditions, customer base, seasonal fluctuations, and so on. These goods can then be sold to the

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final consumers, retailers will in turn market these products to the final consumers, or even back to the farmers themselves. It's important to note that when you're dealing with consumers directly or through retailers, there may be several safety regulations and quality certifications that must be met.

It also involves the treatment of farm waste products. By allowing farm waste to be processed, farmers can safely dispose of potentially hazardous materials as well as increase their profits if these wastes can be sold to processing facilities.

It's important to maintain a good working relationship with the farms from which you source your raw materials or waste and ensure that you are certified to handle these materials or waste. This includes legal certification, health permits, and other important documents. Traceability is an important part of this industry, as inputs are typically closely monitored throughout all stages of the supply chain, including processing.

Since the price of a product usually increases with every step taken along the production chain, it may be most effective to cover several steps in the production process to maximize your profit. Also, logistics are very important in this industry as the costs of procuring, transporting, and processing these raw materials must be carefully weighed, especially since the source of most of these inputs are in rural areas, but consumers stay in predominantly urban areas.

OFFLINE RETAIL AND SALES

Offline retail and sales within the context of agriculture and Agritech usually involve a fixed physical storefront, known as a brick-and-mortar business model. This is different from online retail using a digital marketplace and comes with its

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own unique set of challenges as well. For example, since they cannot leverage the nearly limitless shelf space offered by a digital marketplace, they must deal with the storage, transportation, inventory, and cataloging of physical goods.

This means having an easily accessible location and enough storage space to safely store your wares. These goods must also be well arranged and conveniently kept in units that can be easily purchased by buyers. These stores must also engage in breaking stock, and stock acquisition both in terms of compatibility, consumer demands, future trends, and experience, as well as holding stock periodically depending on seasonal or regular demand.

This means that the store can adopt several methods with regard to product acquisition, storage, and distribution:

- Only ordering these goods when they have been requested by the seller and paid for;
- Order the goods, but only pay the manufacturer when the products have been sold, thereby maintaining a good sales profit margin and also reducing upfront costs to the retailer;
- Obtains the goods from the manufacturer by paying the acquisition costs upfront and then selling them at a retail price to customers.

For such a business model to operate smoothly there must be reliable, efficient, and cost-effective supply chain and storage logistics as such businesses usually run on slim profit margins. This can be achieved through in-depth logistics planning, prudent financial management, and the maintenance of crucial partnerships which function as key supports to the business.

This business does not have to be entirely offline and may

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also have an online component as well, increasing its sales channels and customer channels. In summary, their services must be tailored to the needs of their consumer to ensure optimal productivity, customer retention, and profit.

SERVICE PROVIDER

Farms require utility services such as electricity, water, and safe waste management just as much as any household or industry, if not more. Therefore, the reliable, safe, and continuous supply of these services is of utmost importance to farmers, which is a need that many utility providers capitalize on.

Some of the key operational activities involved in this business model include service provision and quality management, maintenance of the various distribution and production infrastructures involved in the business, and customer management such as scheduling, billing, customer support, and revenue collection. It also includes administrative tasks, planning, logistics, consumption forecasting, and so on.

Such businesses must invest heavily in hiring qualified staff through their Human Resources department and the maintenance of key partnerships, as the continuity and efficiency of essential services form the backbone of this business model. Building a high level of trust requires significant investment into both service provision and customer relations, but such clients usually provide a long lifetime value and significant income, which often balances the high cost of customer acquisition.

The farms which may require these services may be large-scale or small-scale farms, which of course come with different needs and unique challenges for the utility provider.

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The ultimate goal of a service provider should be to offer optimal services to all clients, irrespective of such classifications. This includes clients who may not be able to afford these services and may require tailored solutions. There are different pricing models which may be used in this business model, but the commonest ones are fixed pricing, consumption-based pricing, and block tariffs.

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COMPARING MODELS

Even though each business is different and has its unique method of generating sustainable revenue, business models can be loosely ranked according to certain generalizations which hold relatively true across different business entities within the business model. In this case, the criteria being considered are the Lifetime Value (LTV) and Customer Acquisition Cost (CAC) of a business model.

LIFETIME VALUE

The Lifetime Value (LTV) of a customer is a term used within the world of business and marketing to estimate the net profit which that particular customer will contribute to your business over the entirety of your business relationship with them. It takes into account both present and potential cash inflow from a customer.

There are various factors that can affect this metric. For instance, a business model built on a long-term subscription model will typically have a larger LTV than one built on a one-time payment system. It also allows the business to have a more predictable revenue stream. Other factors which may affect LTV include the brand image, customer relationship with the business, customer satisfaction and commitment, special loyalty benefits, quality of service, and so on.

On that note, some business models such as peer-to-peer lending typically have higher LTV because these farmers service these debts over long periods of time. Mechanization and engineering business models also have relatively high

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LTV because most customers need their services for routine maintenance, troubleshooting, and repairs due to the technical nature of the business.

So do rental and lease business models, as their services are typically required every planting season. Farm advisory services and traceability companies usually have low LTV value because the information they provide per farmer is typically cheap and their business relationships with their clients are mostly short-term.

CUSTOMER ACQUISITION COST

Cost of Acquisition (CAC) refers to the total cost incurred by a business entity in the process of acquiring a new customer. For a business model to be sustainable and turn a profit, the CAC must be lower than the LTV. In simple terms, the cash inflow received from a customer should be higher than the money spent on acquiring the customer.

Similar to LTV, several factors can affect your CAC as well. This includes the revenue spent on marketing and advertising, the number of customers your model requires, the demand for your services, the accessibility of your services to your customers, and so on.

There are several important business strategies that Agritech companies can adopt so that they can reduce their CAC. These include prioritizing your customer base, re-accessing your customer base, adopting the use of affiliate programs, improving customer retention through efficient services and loyalty bonuses, and so on.

Among the various business models in Agritech P2P lending platforms and digital marketplaces typically have low Customer Acquisition Costs due to the advantages their

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business model enjoys from increased digitization. Traceability may also have a low CAC due to the fact that most firms within this industry only need a relatively low number of clients.

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STEPS TO SUCCESS AGRITECH START-UPS

Here are some key steps that every Agritech startup should pay attention to.

IDENTIFY YOUR TARGET MARKET

One of the first steps in building any business model is to identify your target market, the challenges they face, and what solutions you can offer them as your value proposition. This involves deciding whether you are primarily a B2B or B2C firm, carrying out market research, and building an effective market strategy.

PRESENT A CLEAR AND COMPELLING VALUE PROPOSITION

You should be unambiguous about the potential benefits your firm offers clients. This can only be done after you have painstakingly mapped out the ecosystem around which the value chain of your clients operates, pointed out the challenges that you can solve for them, and provided compelling evidence through the use of proof of concepts, past successful projects, and strong data.

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YOU HAVE TO GIVE IT TIME

There are many variables that exist within the agricultural sector, and mapping all these out to create a sustainable business model may take months and even years. Therefore, entrepreneurs within the Agritech industry should understand that building these business models is a marathon, not a sprint and careful deliberation, consistency, and patience are required.

BE SURE YOU KNOW HOW TO SCALE THE TECHNOLOGY

Scalability is quite important within the world of Agritech as farms typically come in a wide range of forms. Therefore, it is imperative that entrepreneurs within the Agritech industry understand the operational activities of their target clients and understand how to adapt their concepts to these unique situations.

DO NOT ASSUME YOU KNOW HOW TO SELL YOUR TARGET'S PRODUCT

Despite the fact that you aim to offer novel solutions to the issues facing the agricultural sector, do not be so quick to assume that you are more well-versed in these issues simply because you are applying new ideas. There is a significant wealth of information available from past business models, and a good entrepreneur knows how to leverage this to improve existing systems and avoid as many mistakes as possible.

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FOCUS ON MORE THAN JUST THE KEY ASSET

While it is important for every startup to be profit- and value-centered, it is also imperative that they understand the ecosystem within which they exist and their position within this value chain. Having a more holistic view of the Agritech industry offers entrepreneurs more opportunities for potential innovation and may offer useful insights as well.

FOCUS ON REGULATORY RISK

Regulation is quite important within the world of Agritech as many of these firms deal with sensitive and sometimes potentially dangerous technology such as genetic modification, pesticides, waste management, and so on. This means that it is very important to pay attention to both the local and international regulations guiding the use of this technology to make sure that all your activities are on the right side of the law.

THE SAME GOES FOR ESG

Environmental, social, and governance (ESG) criteria are a set of guidelines used to screen corporate policies and to encourage companies to act responsibly. With the increased drive for sustainability practices and carbon neutrality, Agritech firms must make sure they adhere to ESG regulations as well.

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CREATE A CROSS-FUNCTIONAL IMPLEMENTATION TEAM

Every startup needs a cross-functional implementation term with a diverse set of skills and different backgrounds which allows them to offer unique and varied services to the business. It's always advisable to seek differing perspectives when dealing with a new business, especially one as innovative as the Agritech industry.

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CONCLUSION

Agritech represents the future, and the future is Agritech. Agritech is far beyond being just another market for investors to make massive profits off, it represents a solution to some of the most important challenges plaguing our society currently. The innovative solutions offered by the Agritech industry do more than just increase efficiency and yield, they may be part of the answer to how to build a sustainable model which will allow humanity to coexist peacefully and prosper with the planet.

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